<u>Last time</u> simplicial complex K

$C_n(K) = \mathbb{Z}/2\mathbb{Z}$ vector space freely gen by the n-simplices in K

$\partial_n : C_n(K)$ to $C_{n-1}(K)$ def by

$\partial_n(\Delta) = \mathrm{sum}_{\text{faces } \Delta'} \Delta'$ for n-simplices Δ

$\partial_n(c + c') = \partial_n(c) + \partial_n(c')$ for all c, c'

mod-2 cycles:

$Z_0(K) = C_0(K)$

$Z_n(K) = \ker(\partial_n : C_n$ to $C_{n-1})$ for $n > 0$

mod-2 boundaries:

$B_n(K) = \mathrm{im}(\partial_{n+1} : C_{n+1}$ to $C_n)$

mod-2 simplicial homology:

$H_n(K) = Z_n(K)/B_n(K)$

[missing fact last time:]

<u>Thm</u> $B_n(K)$ sub $Z_n(K)$ for all n

i.e., if b in $\mathrm{im}(\partial_{n+1})$,
then b in $\ker(\partial_n)$

i.e., if $b = \partial_{n+1}(c)$ for some c in C_{n+1}
then $\partial_n(b) = 0$

i.e., if c in C_{n+1},
then $\partial_n(\partial_{n+1}(c)) = 0$

i.e., $\partial_n \circ \partial_{n+1}$ is the zero map for all n

Ex $\Delta = [a, b, c]$

$\partial_2(\Delta) = [a, b] + [a, c] + [b, c]$

$\partial_1([a, b] + [a, c] + [b, c])$
$= \partial_1([a, b]) + \partial_1([a, c]) + \partial_1([b, c])$
$= ([a] + [b]) + ([a] + [c]) + ([b] + [c])$
$= 0$

Pf of Thm $\Delta = [v_0, v_1, \ldots, v_n]$

will show $\partial_{n-1}(\partial_n(\Delta)) = 0$

will write $\hat{v}_i$ to mean we omit v_i:
e.g., $[v_1, \hat{v}_2, v_3] = [v_1, v_3]$

$\partial_n(\Delta) = \sum_i [v_0, \ldots, \hat{v}_i, \ldots, v_n]$

$\partial_{n-1}([v_0, \ldots, \hat{v}_i, \ldots, v_n])$
$= \sum_{j < i} [v_0, \ldots, \hat{v}_j, \ldots, \hat{v}_i, \ldots, v_n]$
$+ \sum_{j > i} [v_0, \ldots, \hat{v}_i, \ldots, \hat{v}_j, \ldots, v_n]$

$\partial_{n-1}(\partial_n(\Delta))$
$= \sum_{i, j \text{ s.t. } j < i}$
$[v_0, \ldots, \hat{v}_j, \ldots, \hat{v}_i, \ldots, v_n]$
$+ \sum_{i, j \text{ s.t. } i < j}$
$[v_0, \ldots, \hat{v}_i, \ldots, \hat{v}_j, \ldots, v_n]$
$= \sum_{i, j \text{ s.t. } i < j}$
$[v_0, \ldots, \hat{v}_i, \ldots, \hat{v}_j, \ldots, v_n]$
$+ \sum_{i, j \text{ s.t. } i < j}$
$[v_0, \ldots, \hat{v}_i, \ldots, \hat{v}_j, \ldots, v_n]$
$= 0$

recall:

$e_n(K)$ = # of n-simplices in K
= size of any minimal generating set for $C_n(K)$
= $\dim_{\mathbb{Z}/2\mathbb{Z}} C_n(K)$

$\chi(K) := e_0(K) - e_1(K) + e_2(K) - \dots$

$b_n(K)$ = size of any minimal generating set for $H_n(K)$
= $\dim_{\mathbb{Z}/2\mathbb{Z}} H_n(K)$

<u>Thm</u> surprisingly,

$\chi(K) = b_0(K) - b_1(K) + b_2(K) - \dots$

<u>Thm</u> $H_n(K)$, unlike $C_n(K)$, only depends on the homotopy equivalence class of $|K|$

think of b_n, resp. H_n, as a "homotopy-invariant" version of e_n, resp. C_n

this is the starting point of "homological algebra"
= algebra underlying homotopy theory

<u>Q</u> can b_n, H_n tell apart a tetrahedron vs a set of two discrete points?

<u>Ex</u> $K = \{[a], [b]\}$

C_0 gen over $\mathbb{Z}/2\mathbb{Z}$ by $[a]$, $[b]$
C_n trivial for $n > 0$

$Z_0 = C_0$, B_0 trivial

Z_n, B_n trivial for $n > 0$

so $b_0 = 2$ and $b_n = 0$ for $n > 0$

Ex simplicial tetrahedron:

K = {[a], [b], [c], [d],
[a, b], [a, c], [a, d], [b, c], [b, d], [c, d],
[a, b, c], [a, b, d], [a, c, d], [b, c, d]}

[but not [a, b, c, d]!]

C_0 gen over $\mathbb{Z}/2\mathbb{Z}$ by [a], [b], [c], [d]

C_1 gen over $\mathbb{Z}/2\mathbb{Z}$ by [a, b], …, [c, d]

C_2 gen over $\mathbb{Z}/2\mathbb{Z}$ by [a, b, c], …, [b, c, d]

$Z_0 = C_0$ B_0 gen by
[a] + [b],
[a] + [c],
[b] + [c],
[b] + [d],
[c] + [d]

Z_1 = ? B_1 gen by
[a, b] + [a, c] + [b, c],
[a, b] + [a, d] + [b, d],
[a, c] + [a, d] + [c, d],
[b, c] + [b, d] + [c, d]

Z_2 = ? B_2 trivial

Prob $Z_1 = B_1$
Z_2 gen by [a, b, c] + [a, b, d] + [a, c, d] + [b, c, d]

thus:

$H_0 = C_0/([a] + [b], \ldots, [c] + [d])$
H_1 trivial
H_2 gen by $[a, b, c] + [a, b, d] + [a, c, d] + [b, c, d]$

$b_0 = 1$
$b_1 = 0$
$b_2 = 1$

$\chi = 2$